Timothy-Joseph Godwin

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Education

University of Birmingham Sep 2023 – present
M.Eng Computer Science with Software Engineering *Birmingham, UK*

- **Weighted Average to Date:** 71.3% (First Class)
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems & Systems Programming, Machine Learning, Neural Computation, Natural Language Processing, Intelligent Software Engineering, Team Project

Kaplan International College, London Jan 2023 – Jun 2023
Foundation Certificate in Science and Engineering *London, UK*

- **Result:** Pass with Distinction (85%)

Experience

London Success Academy Sep 2025 – Oct 2025
Data Engineering Intern *London, UK*

- Built an ETL pipeline processing 300,000+ records with Python and Apache Spark on a 3-node GCP Dataproc cluster, achieving 80% storage optimisation through Parquet with real-time API integration
- Orchestrated production data workflows using Apache Airflow on Google Cloud Composer with automated scheduling, BigQuery metadata tracking, and data validation ensuring 99%+ pipeline reliability
- Designed BigQuery data warehouse with partitioned/clustered tables processing $15.3M transaction volume across 5 currencies, implementing query optimisation reducing costs by 80% through partition pruning

Uptrail Jul 2025 – Aug 2025
Data Analysis Intern *London, UK*

- Cleaned and analysed 2,992 sales records across 499 customers to identify top loyalty tiers and delivery optimisation opportunities (41% delay reduction potential)
- Developed Scikit-learn logistic regression model on 1,494 streaming customers achieving 76.6% accuracy, identifying mobile usage and geographic patterns as key churn drivers for targeted retention strategies
- Created interactive Tableau dashboard integrating 3 datasets (15,822 records) with 6 calculated KPIs, geographic mapping and customer segmentation analysis

Projects

RAG Study Tool *Python, FastAPI, Claude API*

- Built a retrieval-augmented generation tool that answers questions using uploaded lecture notes, with sources shown for each answer
- Implemented PDF chunking and semantic search using sentence-transformer embeddings stored in ChromaDB
- Added a similarity threshold so the tool declines to answer when no relevant content is found, reducing hallucinations

Campus Events Web Application *JHipster, Angular, PostgreSQL*

- Built a full-stack platform for students to discover and review campus events and societies, as one of 7 developers working in Agile sprints
- Set up the team's CI/CD pipeline with GitHub Actions, Docker and Azure, automating deployment on every merge
- Built REST endpoints in Spring Boot and fixed schema migration conflicts using Liquibase

Technical Skills

Programming Languages: Python, SQL, Java, JavaScript, TypeScript, C, PowerShell

Data Engineering & ML: PySpark (Apache Spark), Pandas, NumPy, Scikit-learn, Tableau, ETL Pipelines

Web Technologies: React, Angular, Spring Boot, HTML/CSS, Bootstrap, Express.js, Node.js, REST APIs

Cloud & Databases: GCP (Storage, Dataproc, Composer, BigQuery), AWS, Docker, GitHub, Azure, Apache Airflow

Additional Information

Languages: English (Fluent), Igbo (Native), Spanish (A1)

Achievements: 2023/24 UG Outstanding Achievement Award (SRM Funded); Silver medallist in state-level Physics and Mathematics Olympiads; Top 3 graduating student & Best Student in Further Mathematics (British Spring College)